**Unit 3: Normalization**

Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity. It involves dividing large tables into smaller, related tables and defining relationships between them.

**First Normal Form (1NF):** A relation is in 1NF if every attribute contains atomic (indivisible) values. No repeating groups or arrays are allowed.

**Second Normal Form (2NF):** A relation is in 2NF if it is in 1NF and every non-key attribute is fully functionally dependent on the entire primary key. Partial dependencies are removed.

**Third Normal Form (3NF):** A relation is in 3NF if it is in 2NF and there is no transitive dependency of non-key attributes on the primary key. In other words, non-key attributes must depend only on the primary key.

**Boyce-Codd Normal Form (BCNF):** A relation is in BCNF if it is in 3NF and every determinant is a candidate key.

**Denormalization** is the process of intentionally introducing redundancy into a database to improve read performance, often used in data warehousing.

**Unit 4: Transaction Management**

A transaction is a logical unit of work that must be executed atomically. The ACID properties are Atomicity, Consistency, Isolation, and Durability. Concurrency control protocols include two-phase locking (2PL) and timestamp ordering. Deadlock occurs when two or more transactions wait indefinitely for each other to release locks.

## Sample Text 2: Banasthali-Specific DBMS Content (for `Banasthali_DBMS.pdf`)

**Banasthali Vidyapith — Department of Computer Science**

**Course: Database Management Systems (CS-301), Semester 5**

**Instructor:** Dr. Meera Sharma

**Course Description:** This course introduces the fundamental concepts of database management systems, including data models, relational algebra, SQL, normalization, transaction processing, and concurrency control.

**Syllabus Coverage:**

- Unit 1: Introduction to DBMS, data models, ER diagrams.
- Unit 2: Relational model, relational algebra, SQL.
- Unit 3: Normalization (1NF, 2NF, 3NF, BCNF).
- Unit 4: Transaction management, ACID properties, concurrency control.
- Unit 5: Recovery techniques, indexing, and query optimization.

**Assessment:**

- Midterm exam: 30%
- Final exam: 50%
- Assignments and lab work: 20%

**Attendance Policy:** Students must maintain at least 75% attendance to be eligible for the final examination. Medical leave requires a certificate submitted within 7 days.

**Recommended Textbook:** "Database System Concepts" by Silberschatz, Korth, and Sudarshan.

**Faculty Contact:** Dr. Meera Sharma, Office: CS Building Room 204, Email: meera.sharma@banasthali.in

**Exam Preparation Tips (from the instructor):** Focus on normalization examples, ACID properties, and SQL queries. Practice ER diagram conversion to relational schemas.